Name:_____ Date:

Incomplete and Codominance – Practice Problems

INCOMPLETE DOMINANCE – a blending of the two alleles appears in the offspring.

1. A **blue-skinned** alien from one quadrant of the universe and a **yellow-skinned** alien from another quadrant of the universe have 4 beautiful alien babies in an intergalactic example of incomplete dominance. Miraculously, the babies all have **green skin** (a blending of yellow and blue). If blue skin is caused by **BB** and yellow skin is caused by **YY**, show how the alien miracle babies were produced (genetically!).

2. In Mountain Mice, the genes for length of tail exhibit "incomplete" dominance. Use a Punnett Square to predict the result of a cross between a homozygous Long-tailed and a homozygous Short-tailed Mountain Boomer. What do the offspring look like?

3. A cross between a 3-eyed alien and a 1-eyed alien produces offspring that have two eyes.

a) If a homozygous 3-eye crosses with a homozygous 1-eye, what is the percent chance of having a two eyed offspring?

b)) If two two-eyed aliens cross, what is the percent chance of having a two eyed offspring?

c)) If a homozygous 3-eye crosses with a 2 -eye, what is the percent chance of having a two eyed offspring?

4. The Four O'clock plant has three genotypes for flower color expressed as three different phenotypes: red flowered plants, white flowered plants and pink flowered plants.

a. Show the expected offspring of a cross between a pink flowered plant and another pink flowered plant.

b. Show the expected offspring between a red flowered plant and a pink flowered plant.

c. Show the expected offspring between a pink flowered plant and a white flowered plant.

CODOMINANCE- both alleles appear in the offspring

5. A cross between a black cat & a tan cat produces a tabby pattern (black & tan fur together).
a. What pattern of inheritence does this illustrate?

b. What percent of kittens would have tan fur if a tabby cat is crossed with a black cat?

c. What percent of kittens would have black fur if a tabby cat is crossed with a black cat?

d. What percent of kitten would have a tabby pattern if a black cat is crossed with a tan cat?

6. In shorthorn cattle hair color is decided by a pair of codominant traits. R is the allele for red hair color and W is the allele or white hair color. A cow with a heterozygous genotype is roan in color, meaning its coat contains both white and red hairs.

a. Describe the expected offspring when a breeder mates cows and bulls of the following phenotypes:

1)red X red

2)white X white

3)red X roan

4)roan X roan

5)white X roan

b. What is the probability of a pair of roan cattle producing (1) a red calf? (2) a roan calf? (3) a white calf?

c. Which would be easier to create heard of all red cattle or a herd of all roan cattle? Explain why!